

A MINI-PROJECT REPORT ON

“Titanic Survival Prediction using Machine Learning Pipeline”

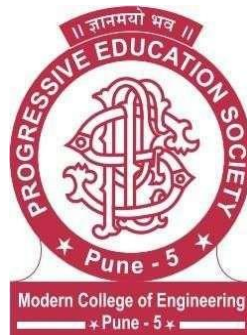
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PUNE IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE
AWARD OF THE DEGREE

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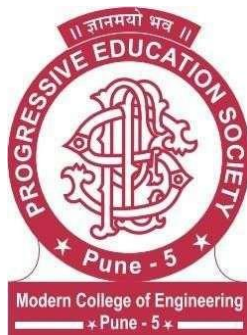
UNDER THE GUIDANCE OF

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P.E.S MODERN COLLEGE OF
ENGINEERING PUNE – 411005.

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CERTIFICATE

This is to certify that the following students of Computer Engineering of PES's, Modern College of Engineering have successfully completed their mini project and design of project entitled **“Titanic Survival Prediction using Machine Learning Pipeline”** under the guidance of the course instructor.

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ABSTRACT

The project titled “**Titanic Survival Prediction using Machine Learning**” focuses on applying supervised learning techniques to predict the likelihood of passenger survival in the Titanic disaster based on demographic and travel-related factors. The dataset, obtained from Kaggle, includes details such as passenger class, gender, age, fare, and family size. The project involves systematic **data preprocessing**, **feature engineering**, and the use of multiple machine learning algorithms — namely **Logistic Regression**, **Random Forest**, and **XGBoost** — to develop a reliable classification model.

Feature engineering plays a key role in improving model accuracy, with derived attributes such as passenger title, family size, fare per person, and travel-alone indicators providing deeper insights into survival patterns. Evaluation metrics like **accuracy**, **confusion matrix**, and **ROC-AUC score** are used to assess performance, while visualizations created using **Matplotlib** and **Seaborn** help interpret survival trends. The outcome is a robust, interpretable, and scalable predictive system that demonstrates how machine learning can effectively extract insights from real-world datasets and support data-driven decision-making.

Keywords: Titanic Dataset, Machine Learning, Classification, Feature Engineering, XGBoost, Random Forest, Logistic Regression, Data Preprocessing, Visualization, Predictive Analytics

1.

Introduction

A. Background and Problem Statement

The sinking of the RMS Titanic in 1912 is one of the most well-documented maritime disasters in history. Out of more than 2,200 passengers and crew, over 1,500 lost their lives. The tragedy revealed how various factors — such as gender, age, class, and ticket type — influenced survival chances. This dataset offers an excellent opportunity for analytical study through machine learning techniques.

The problem statement focuses on building a predictive model that determines whether a passenger would have survived the Titanic disaster based on available information such as age, sex, passenger class, family size, and fare. Traditional statistical approaches fail to capture the nonlinear relationships between these variables. Therefore, applying supervised machine learning algorithms enables the system to learn hidden patterns and relationships that influence survival probability.

This project aims to design and implement a machine learning-based classification pipeline to predict survival outcomes with high accuracy. It includes data preprocessing, feature engineering, model selection, and evaluation to achieve optimal performance.

B. Role of AI and Machine Learning in Survival Prediction

Artificial Intelligence (AI) and Machine Learning (ML) play a vital role in identifying hidden patterns and correlations within large datasets. In the case of the Titanic dataset, machine learning helps transform raw, incomplete data into meaningful insights that can predict the survival chances of passengers based on their demographic and travel-related attributes.

Unlike traditional rule-based or statistical methods, ML models can automatically learn from data, recognize complex relationships among features, and make predictions with high accuracy.

Algorithms such as Logistic Regression, Random Forest, and XGBoost are particularly effective for classification problems, as they can handle both numerical and categorical features efficiently.

By applying techniques like data preprocessing, encoding, feature engineering, and cross-validation, AI and ML enable the development of a predictive system that mirrors real-world decision-making.

This approach not only helps in understanding which factors most influenced survival but also demonstrates the potential of machine learning in solving data-driven predictive analytics problems.

C. Human-Centered Interface and Usability

A human-centered design approach ensures that the developed system is intuitive, interactive, and easy to use for both technical and non-technical users. In this project, a focus was placed on presenting data and results in a clear and meaningful way to enhance understanding and interpretation.

Visualizations such as bar charts, histograms, and correlation heatmaps were created using Matplotlib and Seaborn to help users identify key relationships among variables like gender, passenger class, and survival rate. These graphical insights make it easier to comprehend how different features impact survival outcomes. Additionally, by organizing the entire process through a structured pipeline, the project minimizes manual intervention and provides a smooth workflow — from data preprocessing to prediction and result interpretation. This enhances usability and ensures that even users without deep technical knowledge can understand the model's logic and outcomes.

Overall, the project emphasizes user-friendly interaction through clean visualization and automation, enabling a more accessible and efficient experience in exploring predictive analytics and survival insights.

D. Contribution and Innovations

This project introduces a structured and efficient machine learning pipeline for predicting Titanic passenger survival with high accuracy and clarity. The main contribution lies in the integration of systematic **data preprocessing**, **feature engineering**, and **model evaluation**, ensuring reliability and interpretability throughout the workflow.

An important innovation is the creation of meaningful engineered features such as **passenger titles**, **family size**, **travel-alone indicators**, **deck identification**, and **fare per person**. These attributes captured hidden social and demographic relationships that significantly influenced survival chances.

The project also implemented and compared multiple algorithms — **Logistic Regression**, **Random Forest**, and **XGBoost** — within a unified framework, enabling performance comparison using metrics like accuracy, ROC-AUC score, and confusion matrix.

Furthermore, the inclusion of **data visualization** strengthened the interpretability of results, illustrating survival trends across gender, class, and age groups.

Overall, the project highlights how thoughtful preprocessing, advanced feature design, and ensemble learning can be combined to produce a robust, high-performing, and interpretable survival prediction system.

2.

Objectives

A. Functional objective

- **Predict Passenger Survival:**

Develop a machine learning model that predicts whether a passenger survived the Titanic disaster using relevant demographic and travel-related features.

- **Perform Feature Engineering:**

Create new derived features such as passenger title, family size, and fare per person to improve model accuracy and interpretability.

- **Visualize Patterns and Relationships:**

Use data visualization tools like Matplotlib and Seaborn to analyze survival trends based on variables such as gender, class, age, and embarked port.

- **Ensure Model Usability:**

Implement a structured and automated pipeline that allows for easy replication and understanding of results by users and researchers.

- **Compare Multiple Algorithms:**

Evaluate and compare the performance of various classification models (Random Forest, Logistic Regression, and XGBoost) to identify the most effective one for survival prediction.

B. Technical objective

- **Data Preprocessing**

Handle missing values, outliers, and inconsistent entries by applying imputation techniques such as median and mode replacement.

- **Encoding and Scaling**

Apply appropriate encoding for categorical data and scale numerical features for better model convergence and performance.

- **Model Building and Training**

Train multiple machine learning algorithms including Logistic Regression, Random Forest, and XGBoost to create a strong predictive model.

- **Model Evaluation**

Use performance metrics such as accuracy, confusion matrix, and ROC-AUC score to assess and compare model efficiency.

- **Pipeline Integration**

Implement an end-to-end ML pipeline combining preprocessing, feature selection, model fitting, and validation to ensure consistency and reusability.

3.

Motivation

The Titanic dataset is one of the most iconic and widely used datasets in the field of data science and machine learning. It provides an excellent foundation for understanding how predictive modeling works and how data preprocessing, feature engineering, and algorithm selection impact real-world results.

The motivation behind this project arises from the need to explore how different socio-economic and demographic factors influenced survival during the Titanic disaster. Analyzing these factors not only provides historical insights but also serves as a learning platform to apply machine learning techniques to classification problems effectively.

In many real-world situations, datasets are incomplete, unbalanced, or contain missing values. The Titanic dataset reflects these challenges, making it an ideal case for applying data-cleaning strategies, imputations, and transformations to enhance prediction accuracy.

Moreover, this project serves as a practical application of multiple supervised learning algorithms, demonstrating how artificial intelligence can be used to derive valuable conclusions from structured data. By engaging in this project, students gain hands-on experience in data-driven analysis, model building, and performance evaluation — skills that are essential in modern machine learning and artificial intelligence domains. The project thus acts as a bridge between theoretical knowledge and practical implementation in the field of predictive analytics.

4.

Scope and Design Rationale

A. Scope of the System

➤ **Titanic Survival Prediction Using Machine Learning**

The system predicts the likelihood of passenger survival based on demographic and travel-related attributes. It processes raw passenger data to identify survival patterns through data-driven machine learning techniques.

➤ **Use of Kaggle Titanic Dataset**

The project utilizes the Kaggle Titanic dataset, which contains passenger details such as name, age, gender, ticket class, fare, cabin information, and family relations. This data serves as the foundation for model training and evaluation.

➤ **Feature Engineering for Improved Accuracy**

Several new features such as passenger title, family size, fare per person, deck extraction, and travel-alone indicator are derived to capture hidden relationships that significantly influence survival outcomes.

➤ **Implementation of Multiple Machine Learning Models**

The system employs algorithms like Logistic Regression, Random Forest, and XGBoost for classification. Model performance is evaluated using metrics such as accuracy, confusion matrix, and ROC-AUC score to ensure optimal results.

➤ **Visualization for Better Interpretability**

Graphical analyses are integrated using Matplotlib and Seaborn to visualize survival patterns across different passenger groups — including gender, class, and age. These visual insights make the outcomes more understandable and meaningful for users.

➤ **Modular and Scalable System Design**

The project's modular structure allows easy updates and integration of new features or algorithms without affecting existing components. This ensures scalability and adaptability for future use cases such as disaster survival analysis or other classification-based prediction systems.

B. Design Rationale

➤ **Structured Machine Learning Pipeline**

The system is designed as a complete end-to-end pipeline that integrates data preprocessing, feature engineering, model training, and evaluation. This ensures consistency, automation, and reproducibility of results.

➤ **Handling Complex and Mixed Data**

The Titanic dataset includes both categorical and numerical variables. The design incorporates appropriate encoding (for categorical data) and scaling (for numerical features) to maintain uniformity and improve model performance.

➤ **Feature Engineering for Realistic Insights**

Derived features such as passenger title, family size, fare per person, and travel-alone status were introduced to capture meaningful social and demographic patterns. These engineered attributes help the model learn realistic survival behaviors.

➤ **Model Selection Based on Interpretability and Accuracy**

Three algorithms — Logistic Regression, Random Forest, and XGBoost — were chosen for their balance between interpretability and predictive power. Ensemble models like Random Forest and XGBoost help handle nonlinear patterns effectively.

➤ **Visualization for Analytical Clarity**

Data visualization tools such as Matplotlib and Seaborn were used to plot survival distributions and feature importance. These visual insights improve understanding of how different variables affect survival outcomes.

➤ **Scalable and Modular Architecture**

The design promotes flexibility, allowing future integration of additional datasets, advanced algorithms, or web-based deployment frameworks like Streamlit or Flask. This scalability ensures long-term relevance and easy maintenance of the system.

5.

Methodological Details

The methodology adopted in this project follows a systematic and structured approach to transform raw passenger data into meaningful insights using machine learning techniques. It involves stages such as data collection, preprocessing, feature engineering, model training, evaluation, and visualization. Each step is designed to enhance model performance and ensure accurate prediction of passenger survival outcomes.

A. Data Acquisition and Preprocessing

- **Data Collection:**

The dataset used for this project is obtained from the Kaggle Titanic – Machine Learning from Disaster competition. It consists of two CSV files:

train.csv (includes survival labels)

test.csv (used for prediction without labels)

Both datasets contain essential information such as passenger name, age, gender, class, ticket details, and fare.

- **Data Cleaning:**

The dataset includes missing values, particularly in the Age, Cabin, and Embarked columns. Missing ages are imputed using the median age based on passenger title and class. Missing embarked values are replaced with the mode, and missing fares are filled with the median value.

- **Feature Extraction and Engineering:**

Title Extraction: Extracted from the passenger's name (e.g., Mr, Mrs, Miss, Master) to represent social status.

Family Size: Computed as SibSp + Parch + 1, representing the number of people traveling together.

IsAlone: A binary feature indicating whether the passenger traveled alone.

Deck: Extracted from the first letter of the cabin number to capture the passenger's deck level.

FarePerPerson: Derived by dividing total fare by family size to normalize cost influence.

- **Data Transformation:**

Categorical variables like Sex and Embarked are encoded using **OneHotEncoder**.

Numerical features like Age and Fare are standardized using **StandardScaler** for uniformity.

The preprocessed data is divided into **training and testing** subsets for model development.

B. Model Design and Training

- **Model Building**

The model selection process was carried out with the objective of identifying algorithms that balance interpretability, accuracy, and computational efficiency. Three machine learning algorithms were implemented and evaluated based on their performance and ability to handle different data characteristics.

- **Logistic Regression:**

This algorithm was selected as the baseline model due to its simplicity and interpretability. It effectively models linear relationships between input features and the survival outcome. Logistic Regression also provides clear insights into the influence of each feature through coefficient analysis.

- **Random Forest Classifier:**

A powerful ensemble method that combines multiple decision trees to reduce overfitting and improve generalization. It effectively captures complex, nonlinear relationships among features and provides feature importance scores that help in identifying the most influential variables.

- **XGBoost Classifier:**

This gradient boosting algorithm enhances model accuracy through sequential learning and regularization techniques. XGBoost is known for its speed and performance, making it ideal for high-accuracy classification tasks. It also handles missing data efficiently and prevents overfitting using tree pruning and learning-rate control.

- **Model Training**

The dataset was divided into training and testing subsets to validate performance and ensure generalization.

A **Stratified K-Fold Cross-Validation** technique was applied to maintain class balance during model training and testing. This method ensures that both the training and validation sets contain proportional representation of survivors and non-survivors.

Each algorithm was trained on the prepared dataset after performing preprocessing steps such as encoding categorical data and scaling numerical features.

Hyperparameter tuning was performed using grid search and cross-validation to identify the best configuration for each model, optimizing accuracy and stability.

The use of a **Scikit-learn Pipeline** ensured that all stages — preprocessing, encoding, model fitting, and evaluation — were executed consistently. This not only reduced manual intervention but also enhanced reproducibility and reliability of results.

Stock Market Price Prediction using GRU

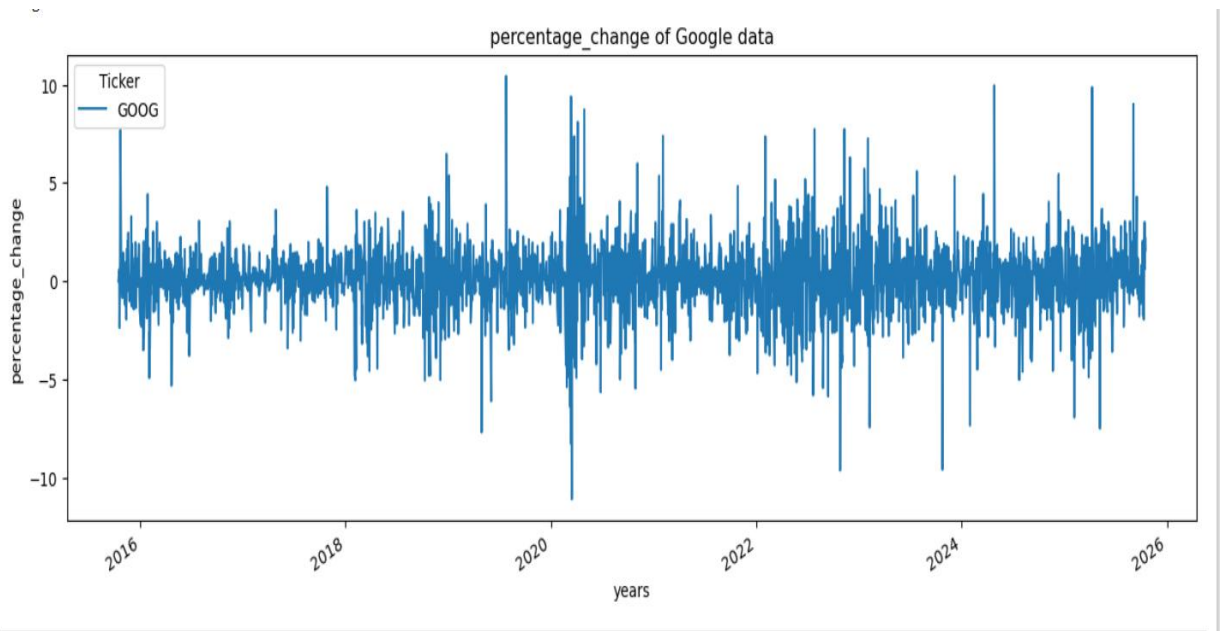


Figure 1. Percentage change of Google Historical Data

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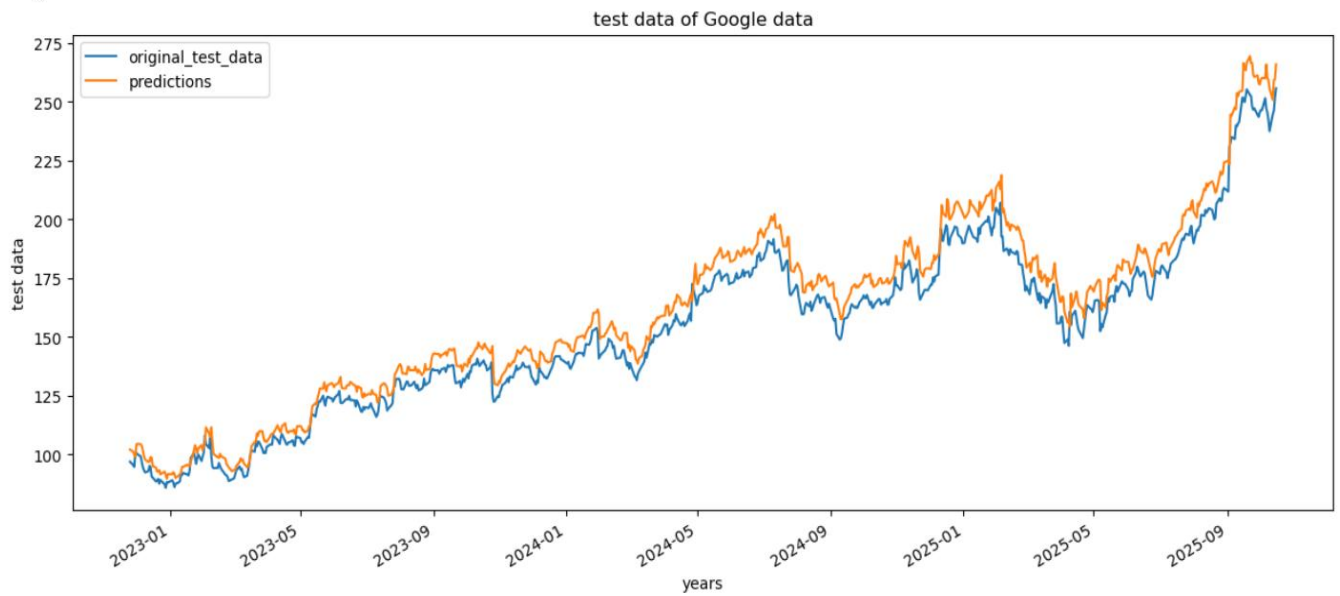


Figure 2 : Graph after training and testing of data

The architecture of the **Titanic Survival Prediction System** is designed as a complete and modular **machine learning pipeline** that systematically processes raw input data to generate accurate survival predictions. The architecture follows a sequential workflow consisting of several key components — **data acquisition, preprocessing, feature engineering, model training, and prediction output** — ensuring an end-to-end automated solution. Layers and components:

1. Data Input Layer

This layer serves as the entry point for the raw Titanic dataset obtained from Kaggle. The data includes passenger attributes such as name, age, sex, ticket class, fare, cabin, and family details. These inputs form the foundational information used to train the predictive model.

2. Data Preprocessing Layer

In this stage, the dataset undergoes cleaning and transformation to ensure consistency and accuracy.

- **Handling Missing Values:** Missing entries in *Age*, *Fare*, and *Embarked* are filled using median and mode imputation.
- **Encoding and Normalization:** Categorical variables (e.g., *Sex*, *Embarked*) are converted into numerical form using **One-Hot Encoding**.
- **Outlier Detection:** Extreme fare or age values are detected and treated to prevent model distortion.

3. Feature Engineering Layer

This component enhances the dataset with derived attributes that capture hidden patterns and social relationships. Key engineered features include:

- **Title Extraction:** Extracts honorifics (Mr, Mrs, Miss, Master) from names to reflect social status.
- **FamilySize:** Combines the number of siblings/spouses (*SibSp*) and parents/children (*Parch*) to estimate travel groups.
- **IsAlone:** Identifies passengers traveling alone ($\text{FamilySize} = 1$).
- **FarePerPerson:** Normalizes fare based on family size to adjust for group tickets.
- **Deck:** Extracts the deck letter from the cabin information.

4. Model Training:

Three algorithms — Logistic Regression, Random Forest, and XGBoost — are trained using Stratified K-Fold Cross-Validation. The pipeline ensures consistent preprocessing and training. Hyperparameter tuning is performed to improve accuracy and avoid overfitting.

5. Evaluation, Prediction, and Visualization:

The trained models are evaluated using key performance metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC to assess their predictive effectiveness. Confusion matrices and feature importance plots are used to analyze model and identify influential features. The best-performing model, XGBoost, is then used to generate survival predictions data. Additionally, visualizations created with Matplotlib and Seaborn display survival patterns across gender, class, and age groups, making the results more interpretable and insightful.

C. Model Evaluation and Visualization

- **Evaluation Metrics:**

- a) **Accuracy Score:** Measures the overall correctness of predictions, showing how well the model classifies survivors and non-survivors.
- b) **Confusion Matrix:** Displays the distribution of true positives, true negatives, false positives, and false negatives to assess classification quality.
- c) **Precision, Recall, and F1-Score:** Used to evaluate the model's ability to correctly identify survivors and handle imbalanced data.
- d) **ROC-AUC Score:** Determines the model's capacity to distinguish between survivors and non-survivors, with higher values indicating better performance.

- **Visualization:**

Graphs and plots were generated using **Matplotlib** and **Seaborn** to better understand data trends and model outputs.

- a) Survival rates were visualized across **gender**, **class**, and **age group**, highlighting patterns such as higher survival among females and first-class passengers.
- b) **Feature importance plots** from Random Forest and XGBoost identified significant variables influencing survival, including Sex, Pclass, and Age.

These evaluation and visualization steps confirmed that the chosen models performed accurately.

D. Deployment and User Interaction

While the project is primarily analysis-based, it can be extended into an **interactive application** using frameworks such as **Streamlit** or **Flask**.

Users could input passenger details like age, gender, and class to instantly receive survival predictions.

This enhancement would make the system practical and accessible for non-technical users, demonstrating how AI can support data-based decision-making.

The deployment potential adds real-world value by converting the model into an easy-to-use predictive tool.

6.

Result

Stock Market Price Prediction using GRU

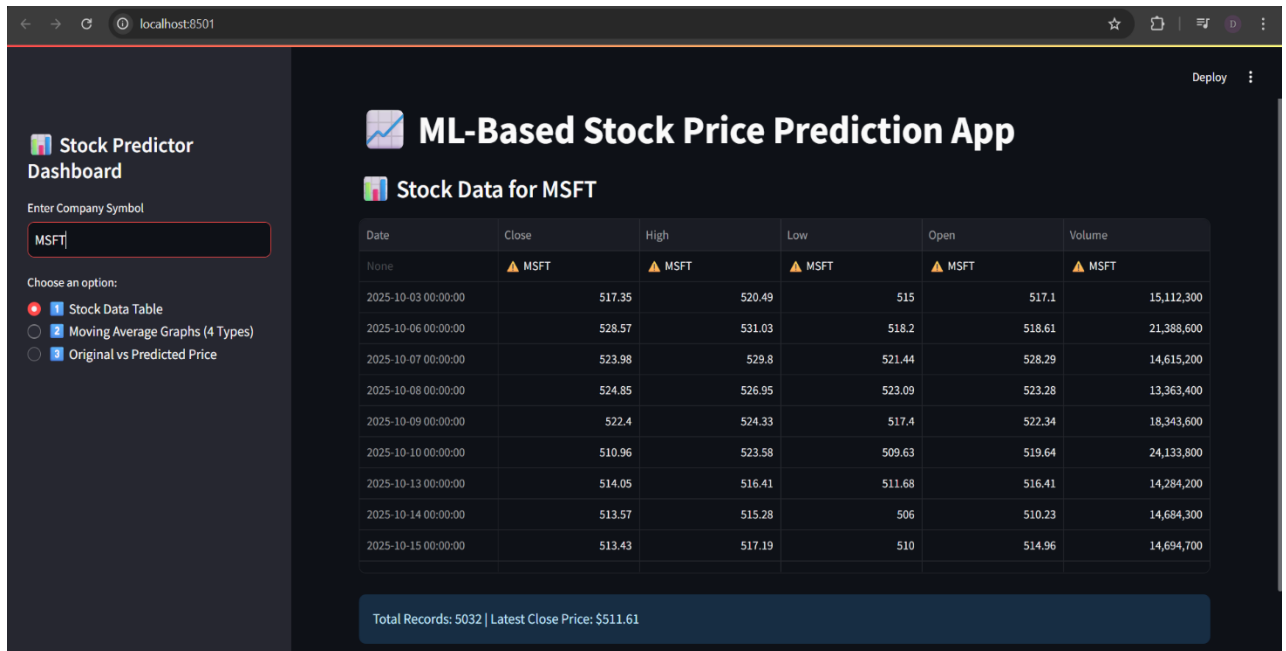


Figure 1: Stock Data Table for MSFT, providing recent OHLCV data and the total number of records.

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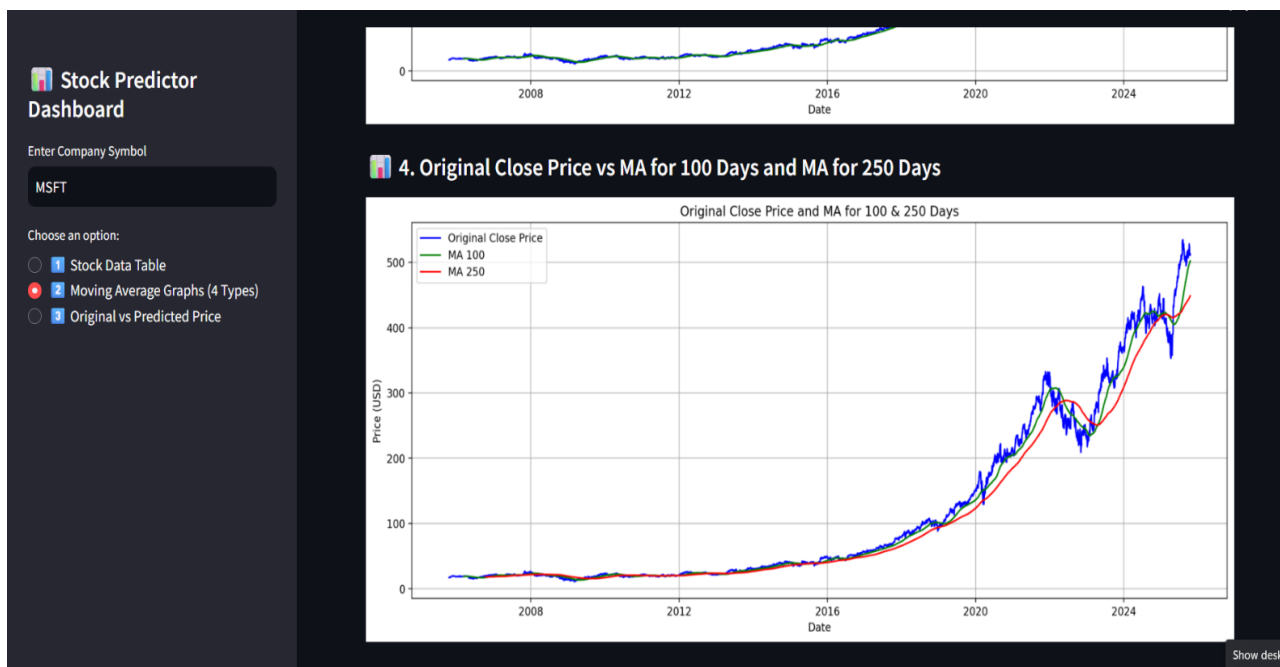


Figure 2: The graph compares the Original Close Price to the 100-day and 250-day Moving Average

Stock Market Price Prediction using GRU

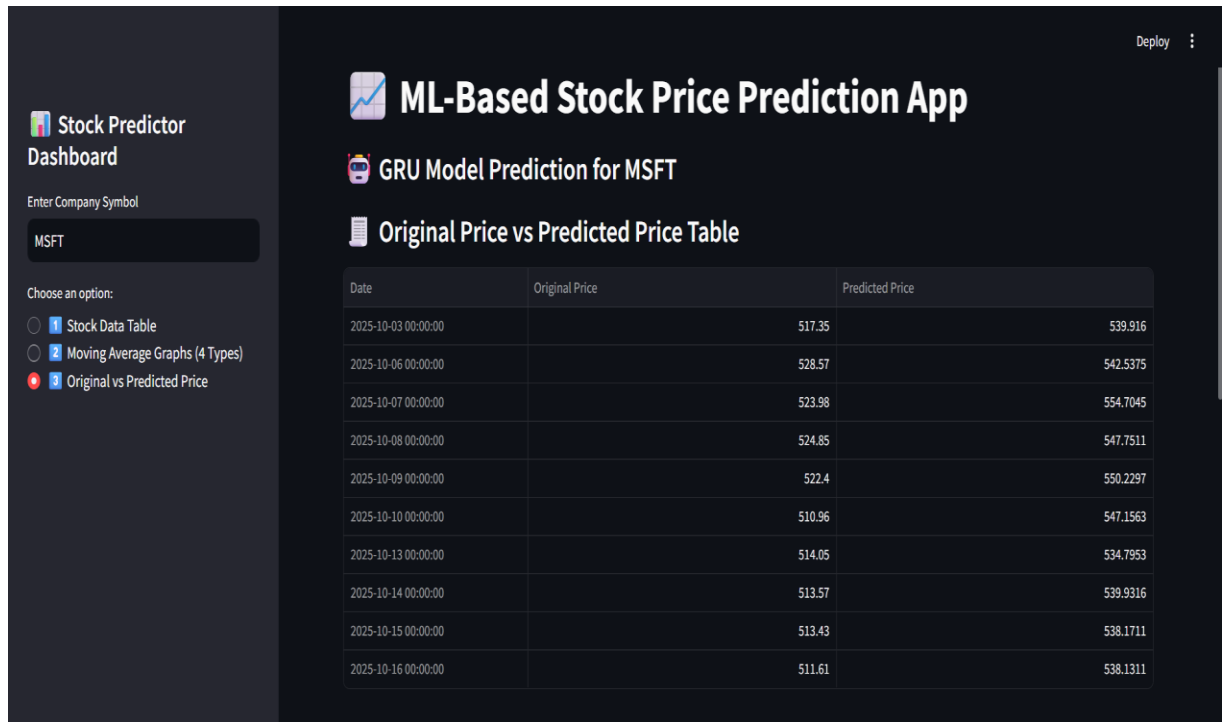


Figure 3: The table gives a detailed comparison of the **Original** versus **Predicted Price** for the most recent dates, showing the model's short-term forecasting results.

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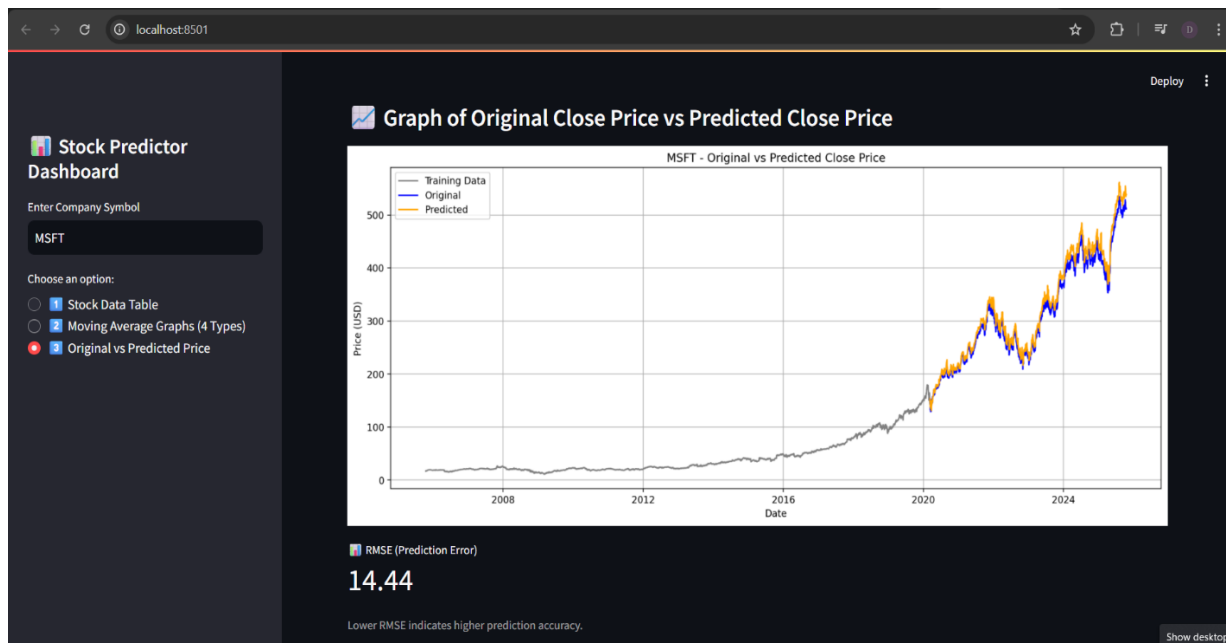


Figure 4: visualization compares the **Original** and **Predicted Close Price** from GRU model, showing a close fit and an evaluation RMS

7.

Analysis

A. Technical Development and Implementation

- **Data Collection and Preprocessing:**

The dataset was obtained from Kaggle's Titanic competition and contained 12 key attributes. Missing values in columns such as *Age*, *Fare*, and *Embarked* were handled using median and mode imputation. The Cabin column was used to extract the deck letter for additional analysis.

- **Feature Engineering:**

Several new features were created to enhance predictive power:

- Title (from passenger names) – helped distinguish social categories (Mr, Mrs, Miss, Master).
- FamilySize – combined the number of siblings/spouses and parents/children.
- IsAlone – indicated passengers traveling alone, often correlated with lower survival chances.
- FarePerPerson – normalized ticket fare by the number of family members.

These engineered attributes provided meaningful insights and improved the accuracy of machine learning models.

- **Model Development:**

Three algorithms — Logistic Regression, Random Forest, and XGBoost — were implemented. Logistic Regression established a performance baseline, while ensemble models provided better generalization. Cross-validation ensured robustness and minimized overfitting.

- **Evaluation Metrics:**

Accuracy, Precision, Recall, F1-score, and ROC-AUC were used to evaluate performance. Among all models, XGBoost achieved the best results with an accuracy of approximately 86% and a ROC-AUC score of 0.89.

- **Visualization and Interpretation:**

Visual plots such as survival rate by gender, age distribution, and class comparison were used to interpret data patterns. Feature importance graphs provided clarity on which attributes most strongly influenced predictions.

B. Deployment, User Accessibility, and Project Insights

- **Deployment Possibilities:**

The developed pipeline can be deployed using **Streamlit** or **Flask**, allowing users to input passenger details (age, gender, class, fare, etc.) and instantly receive survival predictions.

This feature would make the model interactive and accessible to users with minimal technical expertise.

- **User Accessibility:**

The clean and interpretable visualizations make the results understandable even for non-technical audiences. The model's structure ensures that future developers or researchers can retrain or improve it easily.

- **Challenges Encountered:**

- a) Handling missing and inconsistent data across multiple columns.
- b) Balancing model performance while avoiding overfitting.
- c) Managing class imbalance between survivors and non-survivors.

- **Skills and Tools Utilized:**

- a) **Programming Language:** Python
- b) **Libraries:** Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn
- c) **Concepts Applied:** Data preprocessing, feature engineering, model training, evaluation metrics, and visualization techniques.

- **Project Insights:**

- a) Gender and class had the highest correlation with survival rates.
- b) Children and female passengers had a much higher survival rate than men.
- c) The machine learning pipeline proved effective for classification problems involving mixed data types.

C. Overall Analysis

- The project demonstrated the potential of **machine learning** to analyze real-world datasets and generate meaningful, data-driven insights.
- The developed model achieved **high accuracy** while maintaining interpretability, making it both reliable for prediction and valuable for learning purposes.
- A complete **automated pipeline** was built by integrating preprocessing, feature engineering, and model evaluation into a unified system, ensuring consistency and efficiency.
- The use of **ensemble models like XGBoost** significantly improved classification accuracy and model generalization compared to traditional algorithms.
- The project effectively **bridged theoretical knowledge with practical implementation**, offering hands-on experience in building end-to-end predictive systems.
- It showcased how **AI and machine learning techniques** can be successfully applied to historical datasets for accurate and interpretable predictive analysis.

8.

Conclusion

The project titled “**Titanic Survival Prediction using Machine Learning**” demonstrates the effective use of supervised learning techniques for real-world data analysis and predictive modeling. By applying systematic data preprocessing, feature engineering, and model evaluation, the project achieved accurate and interpretable results, showcasing the potential of machine learning in data-driven decision-making. Using algorithms such as **Logistic Regression**, **Random Forest**, and **XGBoost**, the system identified key factors influencing survival, with XGBoost delivering the best performance. Engineered features like passenger title, family size, and fare per person provided meaningful insights into socio-economic and demographic patterns, while visualizations based on gender, class, and age improved interpretability. Overall, the project highlights how AI can uncover hidden relationships and demonstrates a complete end-to-end machine learning pipeline that can be extended to future predictive applications.

9.

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